Circles

Geometry · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10 · worksheet

A+ Maths Hub

Name	Class	Date

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Suggested time: 107 minutes · Total marks: 107

KEY RULES

Circumference	$C = \pi d = 2\pi r$
Area	$A = \pi r^2$
Arc length	$\frac{\theta}{360} \times 2\pi r$
Sector area	$\frac{\theta}{360} \times \pi r^2$
Chord	The perpendicular from the centre bisects the chord

WORKED EXAMPLE 1

[2 marks]

A circle has radius 4 cm. Work out its area to 3 significant figures.

Step 1 Area = πr^2 — $\pi \times 4^2 = 50.265\ldots$

Answer: 50.3 cm^2

WORKED EXAMPLE 2

[3 marks]

A circle has area 49π cm^2. Work out its circumference.

Step 1 Use area to find r — $\pi r^2 = 49\pi, \quad r = 7$

Step 2 Circumference — $2\pi \times 7 = 14\pi = 43.98\ldots$

Answer: 44.0 **cm**

WORKED EXAMPLE 3

[3 marks]

Work out the arc length of a sector with radius 9 cm and angle 40°.

Step 1 Fraction of the circle — $\frac{40}{360} = \frac{1}{9}$

Step 2 Arc = fraction × $2\pi r$ — $\frac{1}{9} \times 2\pi \times 9 = 2\pi$

Answer: 6.28 **cm**

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1 A circle has area 50 cm^2.

Work out its radius, to 3 significant figures.

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

2 A circle has circumference 30 cm.

Work out its radius, to 3 significant figures.

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

3 The radius of a circle is multiplied by 3.

By what number is the area multiplied?

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

4 A circle has radius 18 cm.

Work out the circumference of the circle.
Give your answer correct to 3 significant figures.

..... cm

(Total for question = 2 marks)

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Circles

5 A circle has radius 6.4 cm

Work out the area of the circle.
Give your answer correct to 3 significant figures.

..... cm^2

(Total for question = 2 marks)

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6 A circle has a diameter of 7.6 cm.
Work out the circumference of the circle.
Give your answer correct to 3 significant figures.

..... cm

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2013 · Question 2

7 A circle has diameter 6.3 cm

Work out the circumference of the circle.
Give your answer correct to one decimal place.

..... cm

(Total for question = 2 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2026 · Question 16

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8 The diagram shows sector OPQ of a circle, centre O

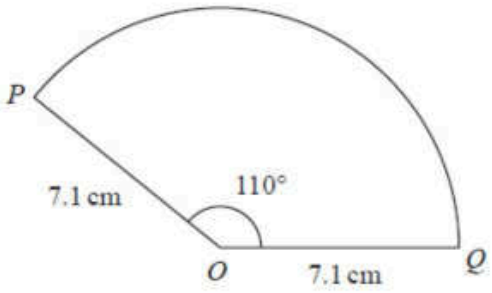


Diagram **NOT** accurately drawn

$OP = OQ = 7.1$ cm
Angle $POQ = 110°$

Calculate the area of sector OPQ
Give your answer correct to one decimal place.

..... cm^2

(Total for question = 2 marks)

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9 $OABC$ is a sector of a circle, centre O

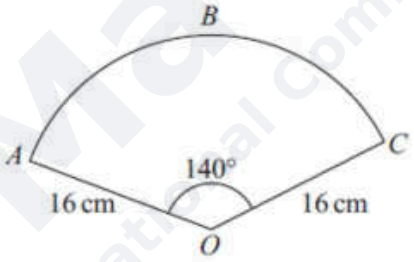


Diagram **NOT** accurately drawn

angle $AOC = 140°$
$OA = OC = 16$ cm

Calculate the area of the sector.
Give your answer correct to 3 significant figures.

..... cm^2

(Total for question = 2 marks)

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10 The diagram shows a sector of a circle with radius 7 cm.

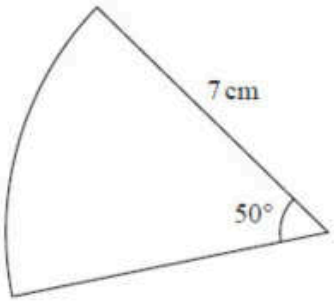


Diagram **NOT** accurately drawn

Work out the length of the arc of the sector.
Give your answer correct to one decimal place.

..... cm

(Total for question = 2 marks)

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11 **15** AOB is a sector of a circle with centre O and radius 12 cm

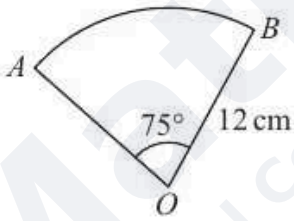


Diagram **NOT** accurately drawn

Work out the area of the sector.
Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 15 is 2 marks)

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12 A circle of radius 5 cm fits exactly inside a square of side 10 cm.

Work out the area of the square not covered by the circle. Give your answer to 3 s

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

13 A rectangle is 12 cm by 8 cm. A circle of diameter 6 cm is cut out of it.

Work out the area that is left, to 3 significant figures.

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

14 A circular pond has radius 4 m. A path 1 m wide goes all the way round it.

Work out the area of the path, to 3 significant figures.

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

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15 **5** The diagram shows a shaded shape made from three identical semicircles, ***X***, ***Y*** and ***Z***

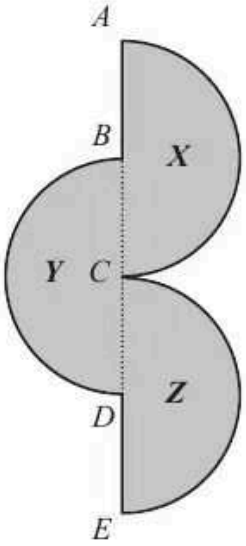


Diagram **NOT** accurately drawn

ABCDE is a straight line.

AC is the diameter of semicircle ***X*** and *B* is the centre of semicircle ***X***
BD is the diameter of semicircle ***Y*** and *C* is the centre of semicircle ***Y***
CE is the diameter of semicircle ***Z*** and *D* is the centre of semicircle ***Z***

$AC = BD = CE = 20$ cm

Work out the perimeter of the shaded shape.
Give your answer correct to the nearest whole number.

..... cm

(Total for Question 5 is 3 marks)

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16 The diagram shows a sector AOB of a circle with centre O

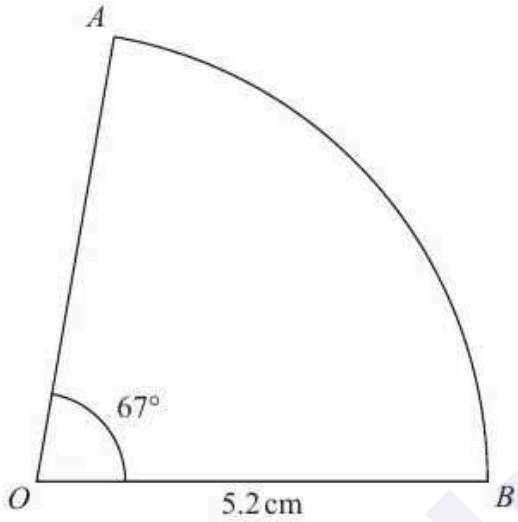


Diagram **NOT** accurately drawn

Angle $AOB = 67°$
$OA = OB = 5.2\,\text{cm}$

Calculate the perimeter of the sector.
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 13 is 3 marks)

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17 OAB is a sector of a circle with centre O and radius r cm.

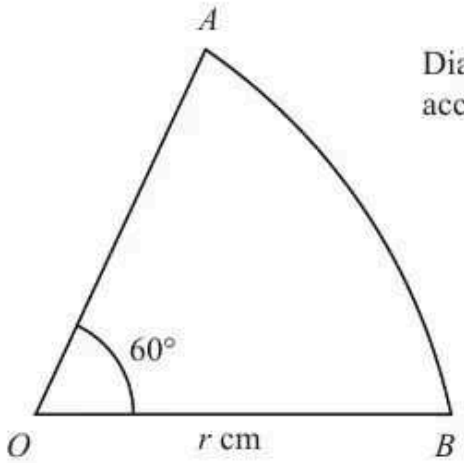


Diagram **NOT** accurately drawn

Angle $AOB = 60°$
The perimeter of the sector is P cm.

Find a formula for P in terms of r
Give your answer in the form $P = r(c\pi + k)$ where c and k are values to be found.

.....

(Total for Question 13 is 3 marks)

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18 A, B and C are points on a circle, centre O

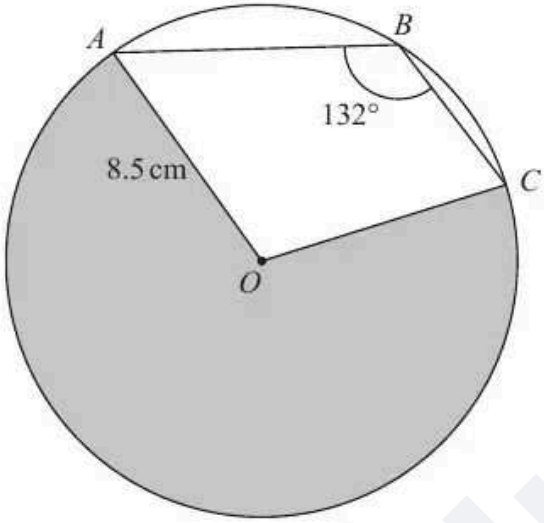


Diagram **NOT** accurately drawn

The radius of the circle is 8.5 cm
Angle $ABC = 132°$

Work out the perimeter of the shaded sector AOC
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 14 is 3 marks)

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19 **6**

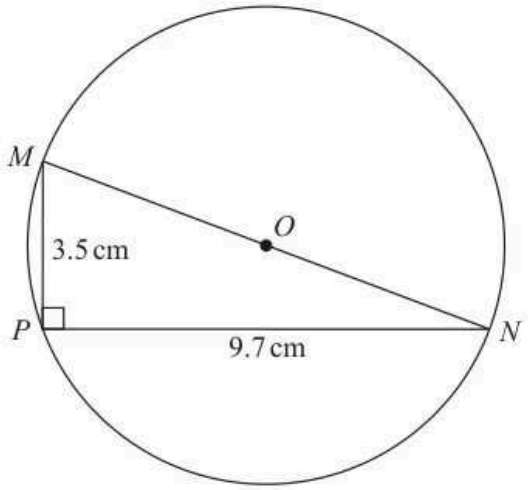


Diagram **NOT** accurately drawn

M, *N* and *P* are points on a circle, centre *O*.
MON is a diameter of the circle.

MP = 3.5 cm
PN = 9.7 cm

Angle *MPN* = 90°

Work out the circumference of the circle.
Give your answer correct to 3 significant figures.

..... cm

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20 A circle centre O has radius 9 cm.

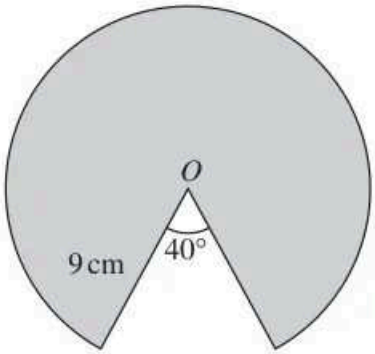


Diagram **NOT** accurately drawn

Calculate the perimeter of the shaded sector of the circle.
Give your answer correct to 3 significant figures.

..... cm

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21 R and T are points on a circle, centre O

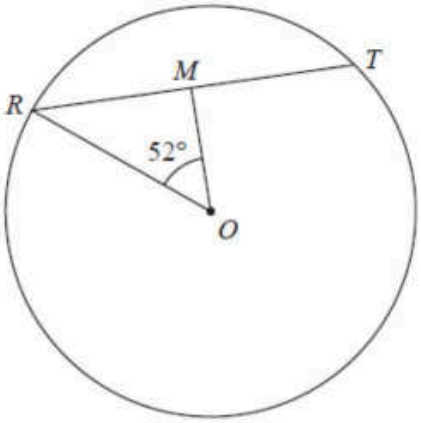


Diagram **NOT** accurately drawn

$RT = 12$ cm
M is the midpoint of RT
Angle $ROM = 52°$

Work out the area of the circle.
Give your answer correct to 3 significant figures.

..... cm^2

(Total for question = 4 marks)

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22 **15** The diagram shows a sector $OABC$ of a circle, centre O

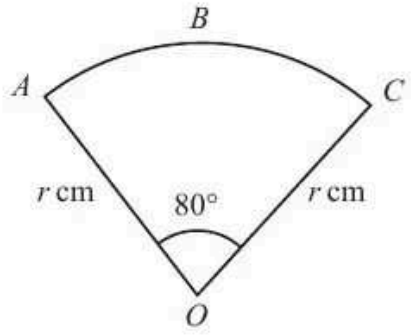


Diagram **NOT** accurately drawn

$OA = OC = r$ cm angle $AOC = 80°$

The area of the sector $OABC$ is 32π cm^2

Work out the perimeter of the sector $OABC$
Give your answer correct to one decimal place.

..... cm

(Total for Question 15 is 4 marks)

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23 *OAPB* is a sector of a circle, centre *O*

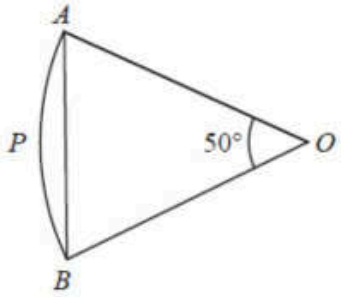


Diagram **NOT** accurately drawn

Angle *AOB* = 50°

Area of triangle *OAB* = 120^2

Work out the area of the sector *OAPB*
Give your answer correct to 3 significant figures.

..... cm^2

(Total for question = 4 marks)

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24 The diagram shows a circle with centre O

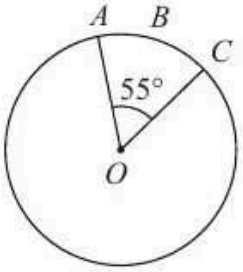


Diagram **NOT** accurately drawn

A, B and C are points on the circle so that the length of the arc ABC is 5 cm.

Given that angle $AOC = 55°$

work out the area of the circle.
Give your answer correct to one decimal place.

..... cm^2

(Total for Question 16 is 4 marks)

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25 **19**

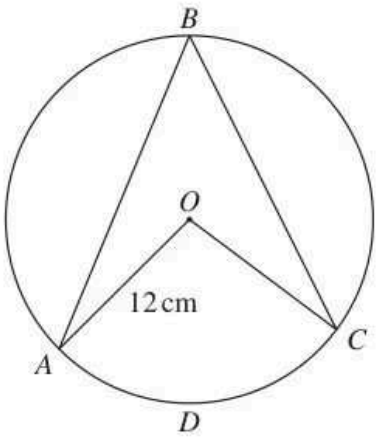


Diagram **NOT** accurately drawn

A, B, C and D are points on a circle with centre O and radius 12 cm.

The area of the sector $OADC$ of the circle is 100 cm^2

Work out the size of angle ABC.
Give your answer correct to 3 significant figures.

.....°

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26 The diagram shows a sector $OABC$ of a circle centre O

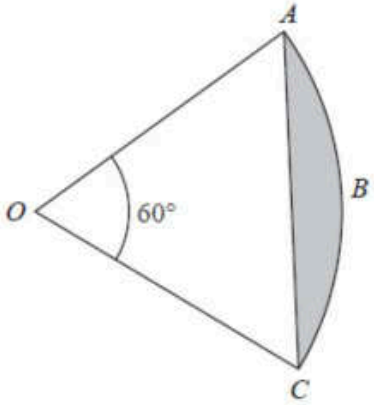


Diagram **NOT** accurately drawn

Angle $AOC = 60°$

The area of the shaded segment ABC is 38 cm^2

Work out the perimeter of the shaded segment ABC
Give your answer correct to one decimal place.

..... cm

(Total for question = 4 marks)

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27 22

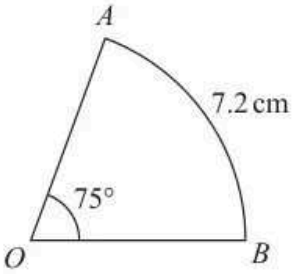


Diagram **NOT** accurately drawn

The diagram shows a sector OAB of a circle, centre O.

Angle $AOB = 75°$
Length of arc $AB = 7.2$ cm

Calculate the area of the sector.
Give your answer correct to 3 significant figures.

..... cm²

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CHALLENGE QUESTIONS

28 The diagram shows a shape made up of three semicircles, enclosing a right-angled triangle.

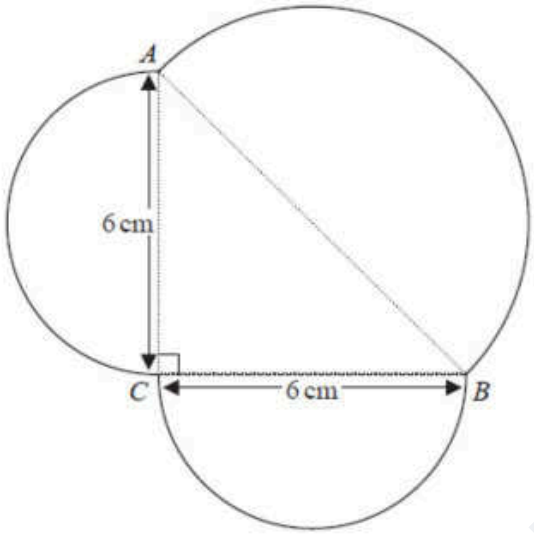


Diagram **NOT** accurately drawn

AB, *BC* and *CA* are each the diameter of a semicircle.

BC = *CA* = 6 cm.

Work out the perimeter of the shape.
Give your answer correct to one decimal place.

..... cm

(Total for question = 5 marks)

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29 **19**

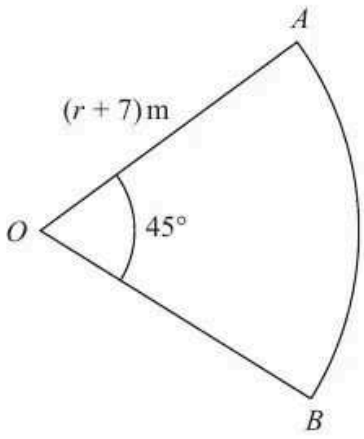


Diagram **NOT** accurately drawn

OAB is a sector **S** of a circle with centre *O* and radius $(r + 7)$ metres.
Angle $AOB = 45°$

A circle **C** has radius $(r - 2)$ metres.

The area of sector **S** is twice the area of circle **C**

Find the value of r
Show your working clearly.

$r =$ .....

(Total for Question 19 is 5 marks)

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30 The diagram shows a sector OBC of a circle with centre O and radius $(6 + x)$ cm.

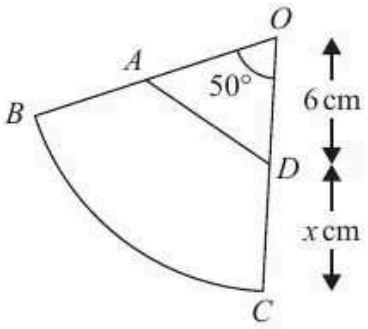


Diagram **NOT** accurately drawn

A is the point on OB and D is the point on OC such that $OA = OD = 6$ cm

Angle $BOC = 50°$

Given that

the perimeter of sector $OBC = 2 \times$ the perimeter of triangle OAD

find the value of x.
Give your answer correct to 3 significant figures.

$x =$ .....

(Total for Question 22 is 6 marks)

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Circles

31 The diagram shows two circles such that the region **R**, shown shaded in the diagram, is the region common to both circles.

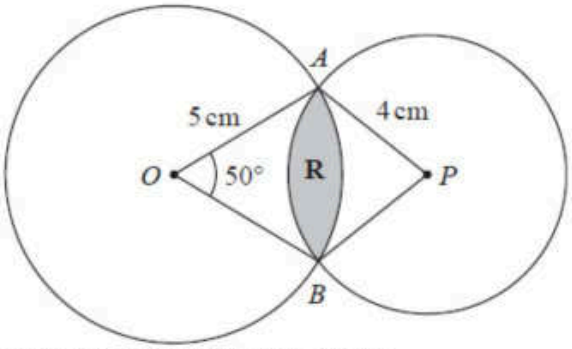


Diagram **NOT** accurately drawn

One of the circles has centre O and radius 5 cm.
The other circle has centre P and radius 4 cm.
Angle $AOB = 50°$

Calculate the area of region **R**.
Give your answer correct to 3 significant figures.

..... cm^2

(Total for question = 6 marks)

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32 **25**

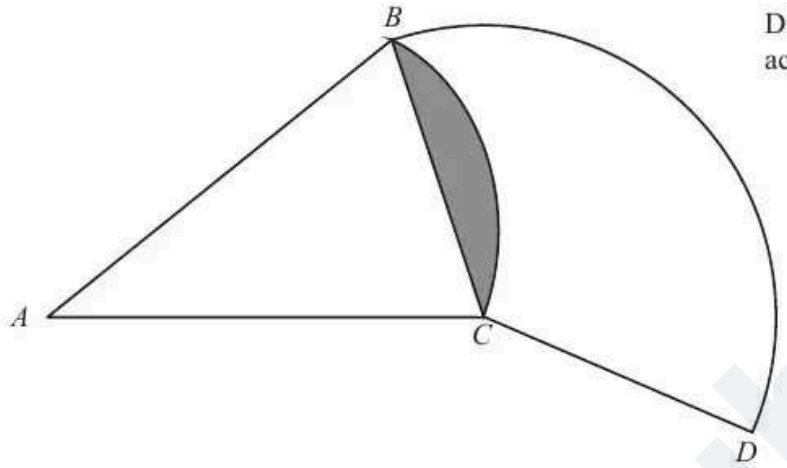


Diagram **NOT** accurately drawn

BAC is a sector of a circle, centre *A*

BCD is a sector of a circle, centre *C*

Angle *BAC* = 40°
Angle *BCD* = 130°
Area of shaded segment = 28 cm^2

Find the length of the arc *BD*
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 25 is 6 marks)

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TOTAL FOR PAPER IS 107 MARKS

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